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■ Lab-5: Build a simple calculator - add, subtract, multiply, divide

■ Lab-5: Build a simple calculator – add, subtract, multiply, divide two numbers

```
# Simple Calculator

num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Choose Operation:")
print("1. Add")
print("2. Subtract")
print("3. Multiply")
print("4. Divide")

choice = input("Enter choice (1/2/3/4): ")

if choice == "1":
    print("Result =", num1 + num2)

elif choice == "2":
    print("Result =", num1 - num2)

elif choice == "3":
    print("Result =", num1 * num2)

elif choice == "4":
    if num2 != 0:
        print("Result =", num1 / num2)
    else:
        print("Error! Division by zero is not allowed.")

else:
    print("Invalid Choice")
```

```
Enter first number: 5
Enter second number: 8
Choose Operation:
1. Add
2. Subtract
3. Multiply
4. Divide
Enter choice (1/2/3/4): 4
Result = 0.625
```

■ Lab-2: Number guessing game + student grade report generator

```
import random

secret_number = random.randint(1, 100)

print("Welcome to Number Guessing Game!")
print("Guess a number between 1 and 100")

while True:
    guess = int(input("Enter your guess: "))

    if guess < secret_number:
        print("Too Low!")
    elif guess > secret_number:
        print("Too High!")
    else:
        print("Congratulations! You guessed correctly.")
        break
```

```
Welcome to Number Guessing Game!
Guess a number between 1 and 100
Enter your guess: 65
Too High!
Enter your guess: 55
Congratulations! You guessed correctly.
```

Lab-3: Student report card system – read CSV, compute grades, write output

```
!PIP install pandas
```

```
/bin/bash: line 1: PIP: command not found
```

```
import pandas as pd

data = {
    "Name": ["Jayanth", "Rahul", "Priya", "Anjali"],
    "Math": [85, 70, 95, 55],
    "Science": [90, 65, 92, 60],
    "English": [88, 75, 98, 58]
}

df = pd.DataFrame(data)

df.to_csv("students.csv", index=False)

print("students.csv created successfully!")
```

```
students.csv created successfully!
```

```
import pandas as pd

def load_csv(filename):
    df = pd.read_csv(filename)
    return df

students_df = load_csv("students.csv")

print(students_df)
```

	Name	Math	Science	English
0	Jayanth	85	90	88
1	Rahul	70	65	75
2	Priya	95	92	98
3	Anjali	55	60	58

```
import pandas as pd

# Load CSV
def load_csv(filename):
    return pd.read_csv(filename)

students_df = load_csv("students.csv")

# Calculate Total and Percentage
students_df["Total"] = (
    students_df["Math"] +
    students_df["Science"] +
    students_df["English"]
)

students_df["Percentage"] = students_df["Total"] / 3

# Grade Function
def calculate_grade(percentage):
    if percentage >= 90:
        return "A+"
    elif percentage >= 80:
        return "A"
    elif percentage >= 70:
        return "B"
    elif percentage >= 60:
        return "C"
    elif percentage >= 50:
        return "D"
    else:
        return "F"

# Apply Grade Function
students_df["Grade"] = students_df["Percentage"].apply(calculate_grade)

# Save Report Card
```

```
students_df.to_csv("report_card.csv", index=False)
```

```
print(students_df)
print("Report Card Generated Successfully!")
```

	Name	Math	Science	English	Total	Percentage	Grade
0	Jayanth	85	90	88	263	87.666667	A
1	Rahul	70	65	75	210	70.000000	B
2	Priya	95	92	98	285	95.000000	A+
3	Anjali	55	60	58	173	57.666667	D

Report Card Generated Successfully!

■ Lab-4: Explore a real dataset (IPL / Zomato / COVID India) – answer 10 questions with Pandas

```
import pandas as pd
```

```
df = pd.read_csv("IPL_matches.csv")
```

```
print(df.head())
print(df.shape)
```

	ID	City	Date	Season	MatchNumber	\
0	1312200	Ahmedabad	2022-05-29	2022	Final	
1	1312199	Ahmedabad	2022-05-27	2022	Qualifier 2	
2	1312198	Kolkata	2022-05-25	2022	Eliminator	
3	1312197	Kolkata	2022-05-24	2022	Qualifier 1	
4	1304116	Mumbai	2022-05-22	2022	70	

	Team1	Team2	\
0	Rajasthan Royals	Gujarat Titans	
1	Royal Challengers Bangalore	Rajasthan Royals	
2	Royal Challengers Bangalore	Lucknow Super Giants	
3	Rajasthan Royals	Gujarat Titans	
4	Sunrisers Hyderabad	Punjab Kings	

	Venue	TossWinner	TossDecision
0	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	bat
1	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	field
2	Eden Gardens, Kolkata	Lucknow Super Giants	field
3	Eden Gardens, Kolkata	Gujarat Titans	field
4	Wankhede Stadium, Mumbai	Sunrisers Hyderabad	bat

(950, 10)

```
import pandas as pd
```

```
print("Shape:", df.shape)
```

```
print("\nTotal Matches")
print(len(df))
```

```
print("\nTotal Seasons")
print(df["Season"].nunique())
```

```
print("\nTop Cities")
print(df["City"].value_counts().head())
```

```
print("\nTop Venues")
print(df["Venue"].value_counts().head())
```

```
print("\nTop Toss Winners")
print(df["TossWinner"].value_counts().head())
```

```
print("\nToss Decisions")
print(df["TossDecision"].value_counts())
```

```
all_teams = pd.concat([df["Team1"], df["Team2"]])
```

```
print("\nMost Matches Played")
print(all_teams.value_counts().head())
```

```
Shape: (950, 10)
```

```
Total Matches
950
```

```
Total Seasons
15
```

Top Cities

City

Mumbai	159
Kolkata	79
Delhi	78
Chennai	67
Bangalore	65

Name: count, dtype: int64

Top Venues

Venue

Eden Gardens	77
Wankhede Stadium	73
M Chinnaswamy Stadium	65
Feroz Shah Kotla	60
Rajiv Gandhi International Stadium, Uppal	49

Name: count, dtype: int64

Top Toss Winners

TossWinner

Mumbai Indians	123
Kolkata Knight Riders	114
Chennai Super Kings	109
Royal Challengers Bangalore	105
Rajasthan Royals	99

Name: count, dtype: int64

Toss Decisions

TossDecision

field	599
bat	351

Name: count, dtype: int64

Most Matches Played

Mumbai Indians	231
Royal Challengers Bangalore	226
Kolkata Knight Riders	223
Chennai Super Kings	208
Rajasthan Royals	192

Name: count, dtype: int64